

Unwitting Default Assumptions: A Pitfall of Categorization

For a quite different example, consider the story of the father who dies in a serious traffic accident, and whose son, a passenger in the same vehicle, is taken by ambulance to the hospital in critical condition. An emergency operation is needed to save his life. The surgeon on duty comes quickly into the operating room, suddenly goes white as a sheet, and exclaims, “I can’t operate on this boy — he’s my son!”

It’s extremely common for people to read and reread this story many times, always bumping into the irreconcilable problem that the boy cannot possibly have two fathers. What is going on? Has the surgeon misperceived the boy’s identity? No. Or was there perhaps not just a biological father but also an adoptive father? No. Did the father somehow get resuscitated and miraculously make it to the hospital before his wounded son? No. In fact, all these questions and answers, despite giving correct information, serve only to further entrench the problem, which is caused by categorical blinders. There is in fact a perfectly reasonable explanation of this situation, which we are confident that every reader, after sufficient reflection, will find.

The manifold ways in which we are continually blindsided by our categories cannot be overemphasized. Take, for instance, the story that you just read about the traffic accident. Did it at any point occur to you that the vehicle involved was a bus? Virtually no one envisions it that way, even though nowhere in the short paragraph recounting the story does it say “car”. It simply never crosses our mind that our initial categorization could have been wrong. The default assumption that comes along with the concept of *traffic accident* is that it involves a *car*, as opposed to, say, a bus, a truck, a motorcycle, a bicycle, a mobile home, and so forth. Such default assumptions are at times very deeply entrenched, and in some cases they are nearly impossible to detect and overthrow. That is why it is so very hard for many readers — even today, when women have made so many strides along the pathway to social equality with men — to figure out how the surgeon’s remark makes perfect sense.

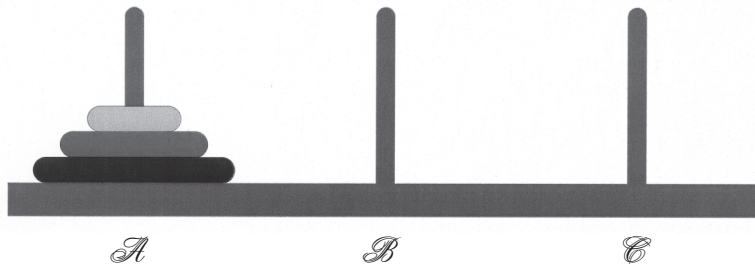
Flitting from One Essence to Another

Categorizations that lead to a mistaken understanding of a situation can have major consequences for someone trying to solve a problem. Indeed, a poor categorization can make a simple problem difficult, if not unsolvable. An external observer might take someone’s being stumped by a simple problem as a symptom of incompetence, when actually the would-be solver is merely trying to solve a problem other than the one that was posed. Just as E.’s biases led her to misconstrue the word “essence” in the lecture title, so the solver has misconstrued the problem because of biased categorization.

Various studies in cognitive psychology have brought out the nature of such mistaken categorizations and are helping to overturn the stereotype that says that if someone fails to solve a problem, it’s only because they didn’t find a successful strategy. That would indeed be the case if the person had perfectly understood the problem statement but didn’t know how to proceed from there. But research has shown that

one of the most frequent sources of trouble in problem-solving is a misunderstanding of the problem statement, meaning that the wrong categories are mobilized. Once the proper categorization is found, finding the solution is often quick and easy.

An example will help make this clear. The study in question involves the famous Towers of Hanoi problem, studied by many psychologists. In a very simple version, there are three disks — one small, one medium-sized, and one large — with holes in their centers, which have been stacked in a pile on one of three posts — post A — and which must be moved to post C while respecting three constraints: (i) only one disk can be moved at a time; (ii) only the uppermost disk in a pile can be moved; and (iii) a larger disk can never be placed on a smaller one.



Many experiments have been done on the topic of how people tackle this challenge and other “isomorphic” challenges (so called because they are in fact just superficial disguises of the same problem, and exactly the same technique will solve them all — for example, the disks can be replaced by acrobats, and the third constraint then says that no acrobat can ever stand on the shoulders of a smaller acrobat). We’ll now shine a spotlight on some interesting results. Cognitive psychologist Jean-François Richard has shown that a significant percentage of elementary-school children, at around age eight, solve this problem with a very large number of moves — around thirty or so — whereas only seven moves will suffice if one knows exactly what one is doing. However, the strategy adopted by these children is anything but random. It seems that they throw in one extra rule that isn’t part of the official statement of the problem. This rule, which they tacitly assume without realizing it, states that one cannot move a disk directly from post A to post C (or vice versa). In other words, these children don’t allow themselves to jump from A to C or the reverse, requiring instead that any disk moved from either A or C must always have B as its landing-spot. If you tackle this alternate challenge yourself, you will soon discover that the addition of this rule slows the solution process up considerably: the minimal solution jumps from 7 moves all the way to 26.

Why would so many children impose this extra rule on themselves, essentially tying one hand behind their back? It all comes down to the concept of *motion*, which seems to be perceived somewhat differently by adults and children. Children often implicitly suppose that moving from one post to another means passing through each intermediate position, whereas adults have a more sophisticated conception of motion, which doesn’t depend on the way one gets from point A to point C. When someone says, “Next Friday I’m going to Arkadelphia”, it could be by car, by bus, by train, or by

plane, and the details of the route make no difference. Adults have no problem distinguishing between a trip as a *state change* (“Today I’m in Bloomington and on Friday I’ll be in Arkadelphia”) and a trip as a *process* (“My Bloomington–Arkadelphia jaunt will carry me through Indianapolis, Little Rock, and several other towns”).

The tendency to see a motion as a *process* rather than as a *state change* is closely related to the idea discussed in Chapter 4, according to which various stages of abstraction of a concept involve stripping away one after another of its less crucial aspects (such as removing from the concept of *desk* the notions of *weight*, *volume*, and *substance*, in the end leaving just the core notion of *workspace*, or removing from the concept of *key* the notions of *metallic*, *long*, *thin*, and *irregular in shape*, in the end leaving just the core notion of *transportable entity that performs unlocking*, such as the magnetic cards that one gets in hotels). It appears from the aforementioned experiments that a good number of eight-year-olds have trouble separating the abstract notion of a *move* from the more concrete notion of the *route* by which the move is accomplished. The former doesn’t care about how it took place and thus allows direct jumps from A to C or back, while the latter insists on passing through all the salient intermediate points. Just as Mica, in the previous chapter, could not help hearing in the word “hump” an earlier collision that had caused the hump, so many young children cannot help hearing in the word “move” the idea of *passing through all intermediate stations*.

Precisely the same phenomenon can be seen in adults when one “dresses” the Towers of Hanoi problem in other “clothing”, thereby yielding one of its isomorphs. Jean-François Richard and his colleague Évelyne Clément studied an isomorph of the Towers of Hanoi in which, rather than moving the disks, one can change their size. In that situation, adults tend to behave exactly as do the children who move the disks. That is, a good fraction of them unconsciously add the extra rule that one can’t change a small disk into a large one (or vice versa) — one has to pass through the intermediate size. In order to jump from size A to size C, they feel they need to take *two* “steps”, stopping momentarily at size B. The cause of this unexpected mental rigidity, which markedly slows these adults up in their solution of the problem, is the same as for the children: they use an overly concrete (and thus naïve) notion of the concept *size change*, not separating the final size from the process or “route” taken in getting to it. For most people, the prototype for the concept *size change* is that of biological growth: one grows from babyhood to adulthood in passing through childhood and adolescence. This model of the concept of *size change* is naïve, though, and adults who get stuck in this trap, unconsciously borrowing the naïve model in tackling the Towers of Hanoi isomorph, are fated to take much longer than those who do not.

It thus appears that one single mental phenomenon is at work both in the “Sources of Essence” trap and in the Towers of Hanoi trap — namely, memories of analogous situations impose themselves without consulting the person who matters — the person in whose head all this is happening. The standard excuse that one gives is along the lines of “I didn’t quite catch what they told me”, or “They sure didn’t make it clear what it was all about”, or “I didn’t pay close attention” — but in fact a more accurate explanation is that one unconsciously succumbed to an inappropriate categorization.